Real-Time Fraud Detection Architecture for Digital Banking: A Systematic Literature Review Applied to the MonedaSegura Case

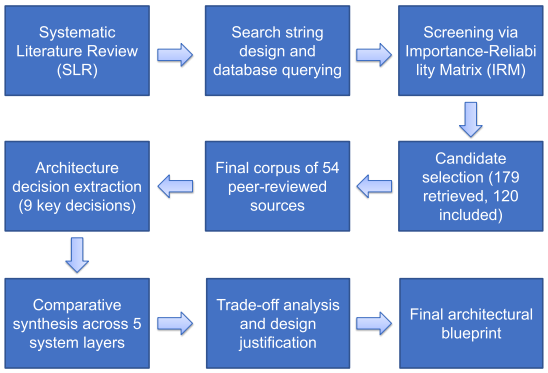
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Summary

Digital banking fraud detection systems face a core tension: rule-based approaches are fast but rigid, while deep learning models are accurate but too slow for real-time transaction scoring. This project designs a fraud detection architecture for MonedaSegura, a digital bank serving 1.8 million users, that resolves this tension through a decoupled offline/online pipeline built on stream processing and a heterogeneous temporal graph neural network (HTGNN).

OBJECTIVE: Design and validate a real-time fraud detection architecture that improves detection accuracy over the current tabular baseline while keeping inference latency within production limits, based on a systematic literature review of 54 peer-reviewed sources.

Methodology



Results

Key comparison: Tabular baseline vs. HTGNN model (Nguyen and Le, ICAART 2025).

Model layer: Kappa + Lambda pattern with BatchNet (offline) and SpeedNet (online) inference.

Additional findings:

Disaggregated state recovery up to 49x faster; checkpoints complete under 3 seconds; prioritized state migration reduces peak latency by nearly 95%.

	AUC-ROC	Avg. Precision	Latency (ms)
Tabular	0.851	0.361	—
HTGNN	0.940	0.641	1.67–1.87

Ingestion and state: Kafka with tiered storage decouples real-time ingestion from long-term retention; Flink 2.0 disaggregated state accepts ~30% CPU overhead in exchange for near-95% faster rescaling recovery.

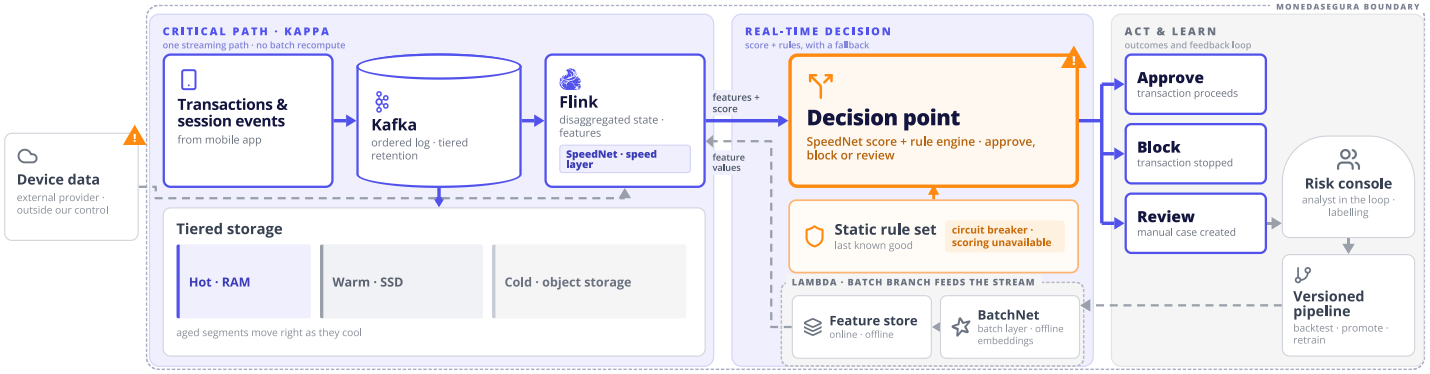
Governance:

A codified MLOps robustness checklist and fully version-controlled tracking (Git-backed) mitigate concept drift and label noise where commercial platforms fall short

Conclusions

- The decoupled five-layer design separates ingestion, state, features, modeling, and governance concerns, allowing each layer to evolve independently.
- The HTGNN model nearly doubles Average Precision (0.641 vs. 0.361) over the tabular baseline while remaining within the transaction-scoring latency budget.
- A Kappa-style architecture was evaluated and documented as a lower-cost interim design for teams without the engineering capacity to operate the full Lambda pipeline.

Architecture



Future Work

Building on the highly secure and production-ready architecture proposed in this work, future research will explore proactive AI-driven resource management. Specifically, integrating reinforcement learning for predictive traffic forecasting can further optimize real-time operator scaling. Additionally, evaluating hyper-efficient checkpointing protocols for disaggregated state storage can minimize infrastructure overhead while preserving absolute transactional integrity and millisecond-level latencies